

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Tech.

Branch : (Electrical Engineering/Electrical &

Power/Electrical & Electronics)

Semester : V

Subject Code & Name: BTEEC502 Microprocessor and Microcontroller

Max Marks: 60

Date: 03/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. Use of non-programmable scientific calculators is allowed.*
- 3. Assume suitable data wherever necessary and mention it clearly.*

	Marks
Q.1 Solve Any Two of the following.	12
A) Draw internal architecture of Microprocessor 8085 with correct arrow notations.	6
B) How is a SIM instruction used to set interrupts? Illustrate with an example.	6
C) Explain following instructions with size, mode of addressing and example:	6
a) LXI H,2500h d) XCHG	
b) LDA 4000H e) DAD H	
c) LHLD 4500H f) ANI 05H	
Q.2 Solve Any Two of the following.	12
A) Interface 2k RAM with μ p 8085 using 1k $\times$ 8 RAMS. Draw address decoding table with start address and last address of a chip.	6
B) Explain the need of de-multiplexing of AD0-AD7 in the memory interfacing with μ p 8085.	6
C) What is DMA? Draw and explain pin configuration of DMA IC 8237 in details.	6
Q.3 Solve Any Two of the following.	12
A) What are the types of digital to analog converter? Explain weighted R-DAC for binary numbers from 000 to 111 with the circuit diagram.	6
B) Write an assembly language program to generate square waveform on CRO using an interfacing IC 8255 with μ p 8085.	6
C) Describe the successive approximation method for A/D conversion.	6
Q.4 Solve Any Two of the following.	12
A) Draw the functional block diagram of micro-controller 8051 and explain each block in brief.	6
B) What is program status word register in μ c 8051? Draw structure of program status word and explain each flag.	6

- C) What are different addressing modes of Microcontroller 8051? Explain with one example. 6

Q. 5 Solve Any Two of the following. 12

- A) Explain following instructions of μ c 8051: 6

i) MOV A,#55H ii) ADD A,R2 iii) ORG 0H iv)
JZ Label v) CPL A vi) MOVC A, @A+DPTR

- B) Draw and explain DC motor controller using a Darlington Transistor. 6

- C) Write an assembly language program to add the contents of RAM locations 60H, 61H and 62H. Store the result in RAM locations 41H (MSB) and 40H (LSB). Assume any suitable input. 6

*** End ***